

Temporal Timber: Time-Encoded Material Passports for Circular Design

Abstract

Contemporary architectural design processes remain largely material-agnostic, assuming a continuous supply of new resources rather than engaging with existing material stock. While circular construction frameworks enable material reuse, they rarely account for temporal variability such as degradation, exposure, and remaining service life. This paper introduces a computational design framework that embeds temporal material information as an operative parameter, aligning the lifespan of individual elements with that of the building as a whole. Using Cross-Laminated Timber (CLT) as a case study, the system generates time-dependent material passports and implements a rule-based substitution model for reallocating elements based on residual capacity and environmental conditions. A demonstration study shows how this approach reduces mismatch between building and material lifespans, enabling more effective reuse of existing components. By reframing design as temporal alignment rather than static specification, the work advances circular design methods and suggests broader implications for lifecycle-based design, maintenance, and material governance.

Keywords: *Material harvesting, circular construction, temporal design, material passports, building-information modeling*

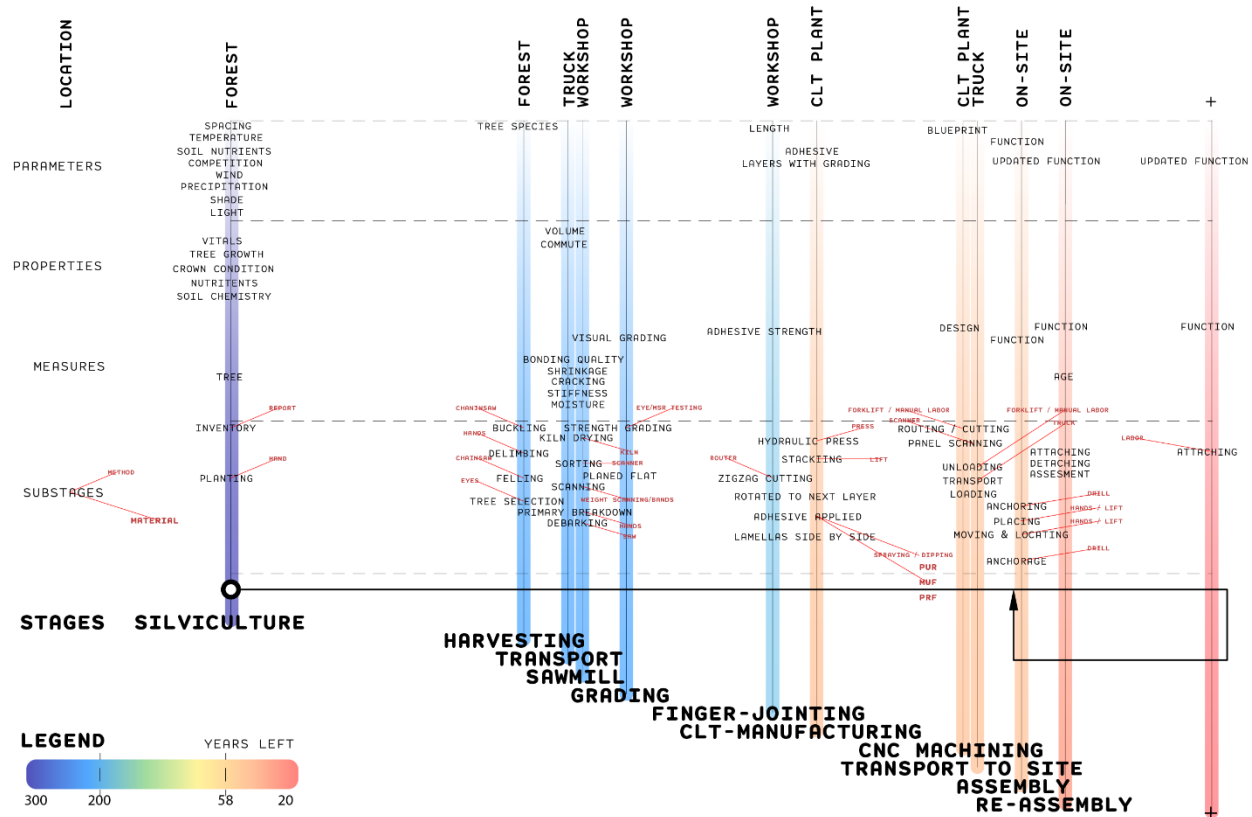
1. Introduction

In a post-extractive future, buildings are increasingly understood as temporary aggregations of materials rather than fixed endpoints, where components persist beyond a single lifecycle and retain value for future use (Geldermans 2016). However, architectural design processes remain largely material-agnostic, assuming a continuous supply of new resources rather than engaging with existing material stock. As a result, design decisions are typically made independently of the availability, condition, and remaining lifespan of materials already in circulation, limiting the capacity to operate within a resource-constrained and reuse-oriented paradigm.

In response, circular construction practices have emerged to support the recovery, reuse, and reintegration of building materials, reframing buildings as material depots or banks (Geldermans 2016; Hebel et al. 2017). Practices such as urban mining extend this logic by treating existing buildings as sources of material extraction, although their implementation remains constrained by gaps in documentation, processing, and reintegration into design workflows (Heisel et al. 2024). Computational approaches have been positioned as critical enablers in this context, offering the capacity to manage the complexity and variability of reclaimed and heterogeneous material stocks. Recent work emphasizes embedding circularity within early design stages through data-driven frameworks that link material information, lifecycle parameters, and decision-making processes (Heisel & McGranahan, 2024). In the context of timber systems, this includes the ability to evaluate and reconfigure elements based on residual capacity, performance constraints, and future reuse potential (Heisel & Bergsagel, 2025).

Concurrent computational approaches, including material passports, BIM-integrated workflows,

and inventory-based design systems, aim to track, classify, and reintroduce materials within early-stage design processes, enabling the integration of heterogeneous and irregular material stock (Ekin & Colakoglu 2023; Batalle Garcia et al. 2021; Augustynowicz et al. 2024). However, despite these advances, such systems largely operate on static representations of materials, limiting their ability to account for temporal variability such as degradation, environmental exposure, and remaining service life. As a result, design processes continue to exclude time-dependent material behavior as an active parameter in decision-making.



A central limitation of current design practice is the misalignment between the lifespan of buildings and the lifespans of their constituent elements. While materials often retain significant

residual capacity, buildings are frequently decommissioned due to functional, economic, or regulatory factors rather than material failure (Berglund-Brown et al. 2025), resulting in the premature loss of usable components. As Brand (1994) describes through the concept of “shearing layers,” building systems operate on distinct temporal cycles, with structure, envelope, and services aging at different rates (Figure 2). However, contemporary design practices continue to treat buildings as temporally unified entities, rarely accounting for these differences. This produces a disconnect between part and whole, where building lifespans are defined independently of the temporal capacities of their materials. Reconciling this relationship requires a design approach in which time is embedded directly within material representation and decision-making.

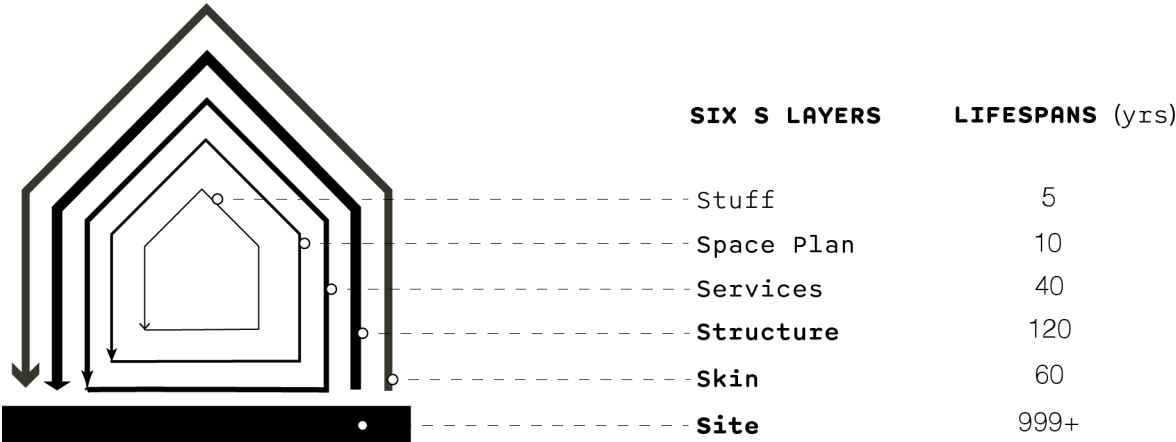


Figure 2. Updated diagram of 6S with average lifespans (Brand, updated by authors)

This paper introduces a computational design framework that embeds temporal material information as an operative parameter within the design process. Using Cross-Laminated Timber (CLT) as a case study, the system integrates lifespan-related attributes—derived from environmental exposure, material properties, and prior use—into element-level material passports, which inform a rule-based substitution model for aligning the lifespan of individual components with that of the building as a whole. By shifting from static material specification to time-informed material allocation, the research reframes design as a process of temporal alignment between parts and wholes. The contributions of this work are: (1) a method for encoding time-dependent attributes within material passports, (2) a substitution framework that aligns material and building lifespans, (3) an interface for inventory-based allocation of temporally differentiated elements, and (4) a demonstration of how temporal mismatch between building and material lifespans can be reduced through targeted substitution.

2. Methods

The proposed system is a computational design framework that operates on a building-scale material inventory, integrating geometric model data with time-dependent material attributes to support lifespan-aware design decisions. Rather than treating materials as static components, the system models each element as a temporally conditioned unit, whose projected lifespan is derived from its material properties, environmental exposure, and prior use. These attributes are encoded within element-level material passports, which serve as the primary data structure for evaluating

and reallocating components within the design. By linking element-specific temporal behavior to building-scale performance requirements, the framework enables the alignment of material lifespans with the intended service life of the building.

The framework operates across two interconnected scales: the individual material element and the building as an assembly. At the element level, temporal attributes are encoded within material passports, allowing each component to be evaluated as a time-dependent unit shaped by exposure, material properties, and prior use. At the building scale, these temporally differentiated elements are aggregated and reallocated through a rule-based substitution system, enabling alignment between the lifespan of individual components and the intended service life of the overall structure. The following sections describe the development of the computational tool, the construction of element-level material passports, and the implementation of the substitution model at the scale of the building.

2.1 Computational Tool

The proposed system operates in two interconnected stages, where model data and material characteristics are combined to construct a material passport that includes lifespan-related attributes.

The system constructs a material passport by integrating geometric model data (including element location, dimensions, and spatial classification) with material attributes (such as species, grade, installation year, and exposure conditions), producing a structured dataset in which each element is associated with time-dependent lifespan parameters. A rule-based classification process organizes elements across hierarchical scales—global (zones, floors), element (wall, floor, façade), and local (relative position within the building)—enabling the assignment of differentiated exposure conditions and lifespan parameters. At the local level, spatial parameters such as distance to center and position within the building are used to assign differentiated lifespan conditions. Lifespan is modeled as a conditional attribute derived from the interaction between spatial exposure, environmental risk factors, and material properties, allowing elements within the same system to exhibit differentiated rates of degradation and remaining service life.

This model-derived dataset is then combined with material attributes, including species, grade, installation year, and design risk conditions, producing a structured dataset in which each element is defined by both its geometric role and material history. Rather than treating lifespan as a predefined value, the system encodes it as a conditional attribute emerging from the interaction between spatial position, use scenario, and material properties.

This data structure enables the design process to operate on a dynamic inventory of temporally differentiated elements, where decisions about material allocation and substitution are informed by the remaining lifespan of each component in relation to building-level requirements. This integration of geometric, material, and temporal data establishes the basis for both element-level evaluation and building-scale substitution, which are developed in the following sections.

2.2 Element: Material Passports

At the element level, each component is represented as a material passport—a structured data object that consolidates geometric, relational, and material attributes with time-dependent

lifespan parameters. The material passport serves as the primary unit of analysis within the system, enabling each element to be evaluated as an independent, temporally conditioned entity.

Each passport includes a unique identifier (element ID, zone ID), spatial classification (zone, system, and position within the building), and geometric properties (dimensions, thickness, and vertex coordinates). These attributes are combined with material characteristics, including species, grade, installation year, and design risk conditions, which together define the baseline performance of the element.

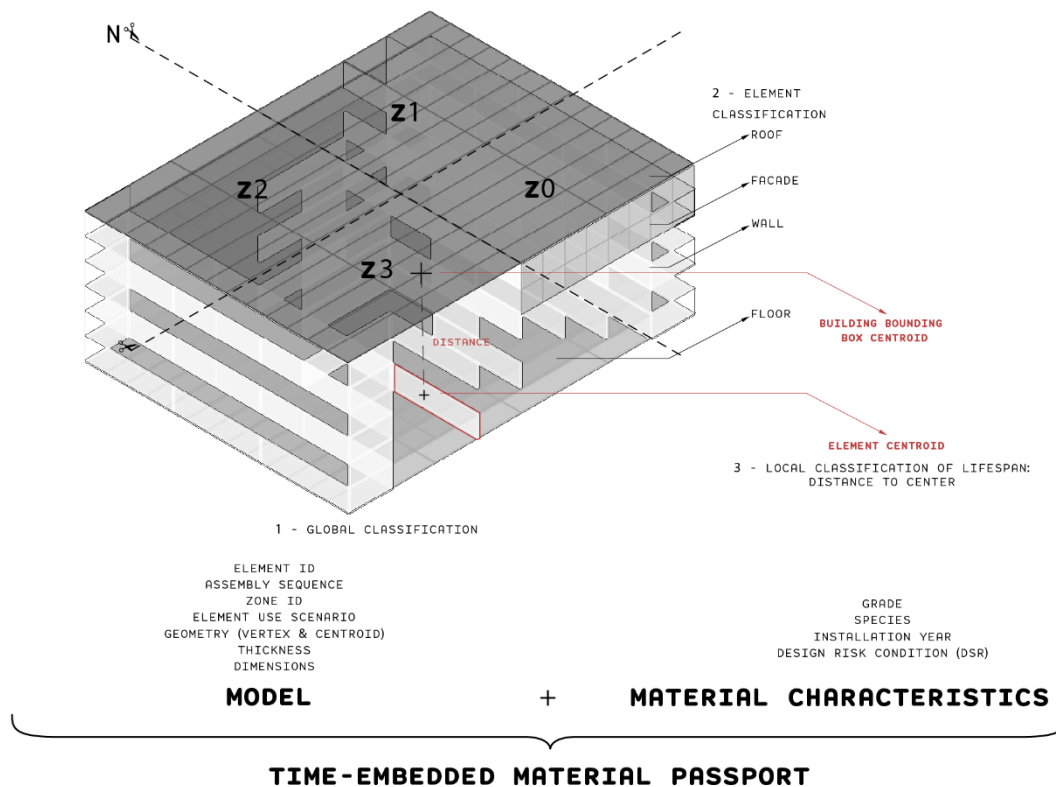


Figure 3. Material passport identification workflow

Lifespan is modeled as a conditional attribute derived from the interaction between material properties, spatial exposure, and environmental risk factors. Rather than assigning a fixed service life, the system calculates differentiated lifespan values across elements based on their position within the building and their exposure to moisture and other degrading conditions. As a result, elements within the same system may exhibit distinct temporal behaviors, reflecting uneven rates of degradation and remaining service life.

This structure enables the system to operate on a dynamic inventory of temporally differentiated components, where each element carries a projected lifespan that informs its suitability for reuse, substitution, or continued deployment. The material passport therefore functions not only as a record of material information, but as a computational representation of element-level temporal capacity within the broader building assembly.

The substitution model is formulated as a multi-objective problem, prioritizing moisture content as the primary variable. This emphasis reflects the role of moisture as a key driver of timber degradation and lifespan, as it directly influences dimensional stability, decay processes, and long-term structural performance in wood-based systems (Forest Products Laboratory, 2010). By foregrounding moisture, the system evaluates element substitutions based on differential exposure conditions and their resulting impact on performance over time (Figure 3).

2.3 Assembly: Web-based Substitution Tool

At the building scale, the system is deployed as a web-based visualization interface that operates on the extracted inventory and enables interactive manipulation of building elements. The interface presents the building as a field of discrete components; each linked to its material passport and visualized with associated temporal attributes such as remaining lifespan and exposure conditions.

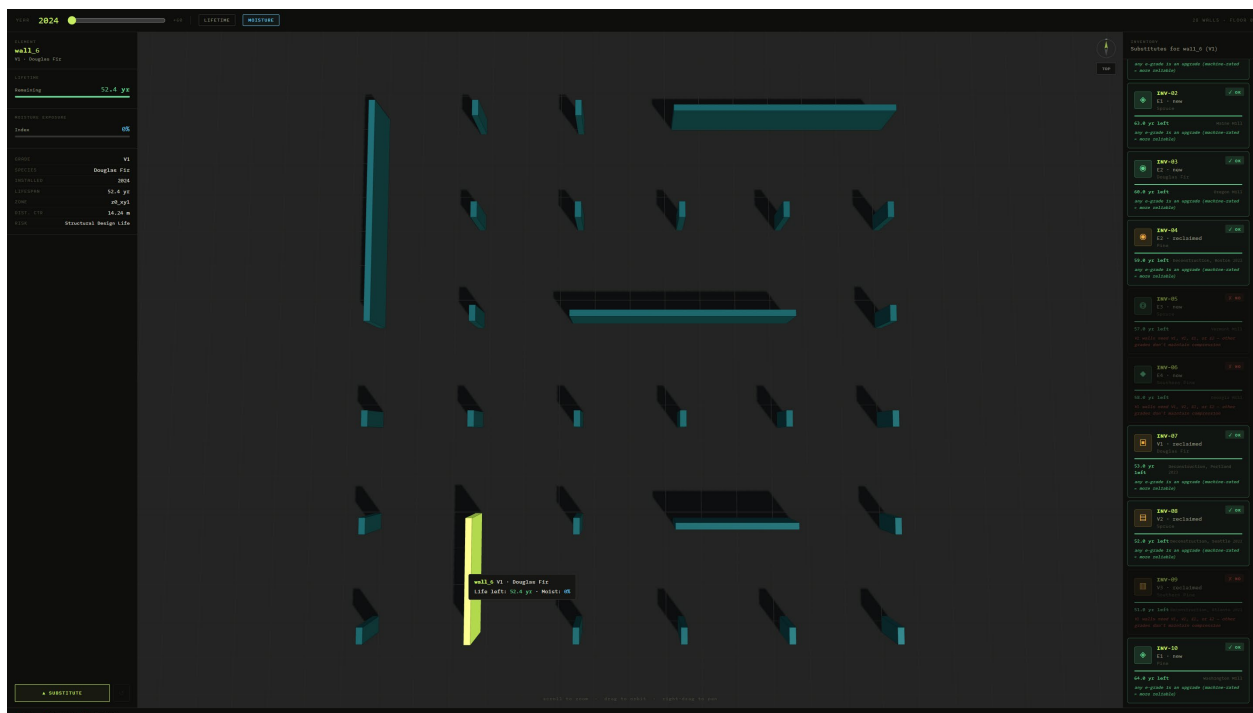


Figure 4. Web-based Substitution Tool User Interface

The UI enables users to isolate elements, inspect their properties, and compare alternative material instances through a substitution panel (Figure 4). Available elements; new, reclaimed, or reconditioned, are presented alongside their respective attributes (e.g., species, condition, remaining life), allowing direct replacement within the building model. This interaction operates in real time, updating both the selected element and its relation to the overall building system.

Rather than representing the building as a fixed assembly, the interface enables the alignment of building-level service life with the residual lifespans of its constituent elements. Through element-level substitution, components with lower remaining service life than the target building lifespan can be replaced with alternatives from the inventory that exhibit higher residual capacity

and more stable moisture conditions. This operation reduces temporal mismatch between system-level and component-level performance by recalibrating the distribution of element lifespans within the building. In doing so, the system minimizes premature obsolescence of materials and extends the effective service life of the building through targeted material reconfiguration.

The temporal behavior of elements is modeled through a rule-based decay framework, where lifespan reduction is calculated as a function of spatial exposure gradients and moisture-driven conditions over time (Figure 5).

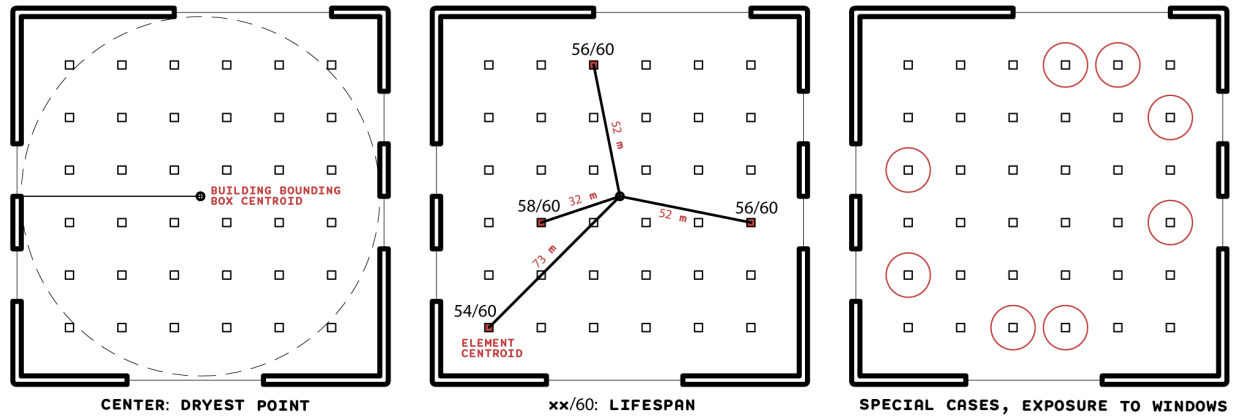


Figure 5. Rules of lifespan decay determination: distance to the dryest spot at t_0

2.4 Temporal Grounding: Replacement Logic and Substitution Matrix

The temporal behavior of the system is operationalized through a rule-based replacement logic and a substitution matrix that governs allowable element exchanges.

2.4.1. Replacement logic

For this model, design intent and functional roles are preserved. Elements are substituted within fixed application categories (e.g., wall, floor, roof), such that a wall remains a wall and is only replaced with elements capable of fulfilling the same functional role. Structural load scenarios are not explicitly recalculated within the model; instead, compliance is ensured through pre-defined grading systems and material classifications that encode performance thresholds.

Replacement is therefore constrained by compatibility rules that maintain or improve required performance criteria (e.g., stiffness for floors, compressive capacity for walls, weight constraints for roofs). This allows substitution to operate within a controlled design space without redefining the structural system at each iteration.

2.4.2. Substitution Matrix

The substitution matrix defines a rule-based decision framework that governs permissible exchanges between elements based on performance equivalence and temporal compatibility. As shown in *Table 1*, acceptable substitutes are defined according to application-specific criteria:

- **Floors and Roofs** require equivalent or greater bending stiffness (EI), restricting substitutions that reduce structural performance
- **Walls** prioritize compressive strength (Fc), allowing substitutions that maintain or enhance load-bearing capacity.
- **Wall bases and high-risk zones** require density and durability thresholds, limiting substitution options to elements with equivalent or superior material properties.

The matrix encodes directional substitution logic (e.g., upgrades allowed, downgrades restricted), ensuring that replacements contribute positively to the temporal performance of the system. For example, elements with higher stiffness or strength may replace lower-performing ones, while the inverse is constrained unless explicitly permitted by the application context.

Table 1: Substitution Matrix

Original	Application	Acceptable Substitutes	Reasoning
E1 (SPF)	Floor/roof	E1, E2	E2 has higher bending strength, same or better stiffness
E1 (SPF)	Wall	E1, E2, E4	Maintain or improve compression
E2 (DF-L)	Floor/roof	E2	E1 is less stiff, less acceptable for floors
E2 (DF-L)	Wall	E2, E4	E4 adds compression capacity
E3 (Northern)	Floor/roof	E3, E1, E2	E1 and E2 are stiffer, upgrade
E3 (Northern)	Wall	E3, E1, E2, E4	All maintain or improve
E4 (Southern Pine)	Wall base	E4 only	Unique density advantage, no true substitute
E4 (Southern Pine)	Floor	E4, E2, E1	Might be too safe but acceptable
E5 (Hem-fir)	Any	E5, E1, E3	E5 is niche, E1 or E3 natural upgrades
V1 (DF-L)	Wall	V1, V2, E1, E2	Any E-grade is upgrade
V2 (SPF)	Wall	V2, V1, E grades	E-grades acceptable as substitution for V
V3 (Southern Pine)	Wall base	V3, E4	Same density requirement as E4 base walls

2.4.3. Design Risk Conditions

In addition to material grading, the model incorporates design risk conditions such as moisture exposure, spatial position, and proximity to high-risk zones. Temporal performance is evaluated across layered lifespan definitions, including product lifespan, structural design life, and actual potential life.

A key paradox emerges in engineered wood systems such as CLT: while timber as a material can

persist for over a century under favorable conditions, its effective lifespan is often constrained by the shorter, guaranteed lifetime of adhesive bonds. As a result, substitution decisions must account not only for structural performance but also for the discrepancy between material longevity and product-level limitations, prioritizing elements that extend building service life beyond these constraints.

- To evaluate the implications of temporal mismatch between building and material systems, the model defines three primary lifespan gap scenarios based on the relationship between **building lifespan (BL)** and **material lifespan (ML)**:
 - **Option 1: BL > ML**
The building outlives its constituent elements. This condition results in premature material obsolescence, requiring replacement cycles within the building's service life. It represents the most critical mismatch, where components must be substituted to maintain overall system performance.
 - **Option 2: BL < ML**
The building is decommissioned before its elements reach the end of their service life. This condition leads to underutilization of material capacity and contributes to material waste, as elements with remaining performance are removed from use.
 - **Option 3: BL $\approx$ ML**
The building lifespan is approximately aligned with the lifespan of its elements. This condition represents an optimized temporal configuration, minimizing both premature replacement and underutilization of materials.

These scenarios can be evaluated with and without **Design Risk Conditions (DRS)**, which introduce variability through exposure factors such as moisture, spatial position, and environmental risk. The inclusion of DRS enables the model to move from idealized lifespan alignment toward more realistic, context-dependent performance conditions (Figure 6).

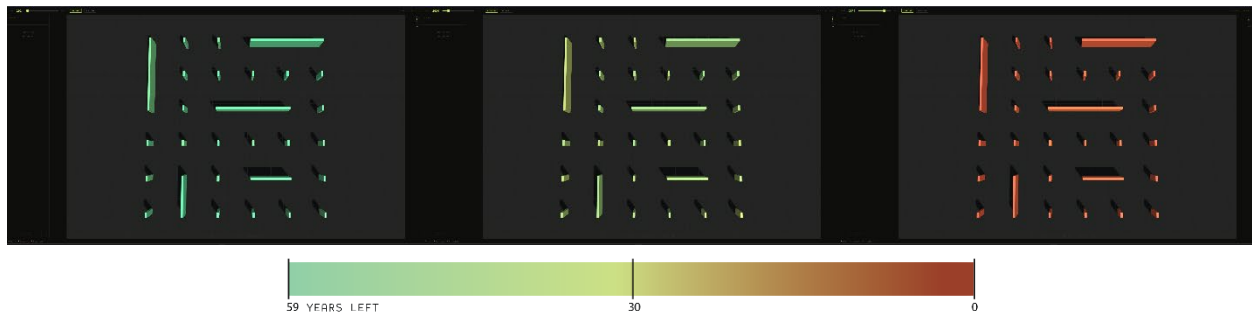


Figure 6. Image taken from the Web-based tool showing lifespan indication of UI model in 2024, 2054, 2084 showcasing the lifespan decay according to placement within building mass.

2.5. Demonstration Study

To demonstrate the operationalization of these scenarios, the proof-of-concept focuses on a **single-story building assessment**, analyzing wall, column, and floor elements under controlled substitution conditions. The study specifically examines the **BL > ML scenario**, where element lifespans fall below the target building lifespan, triggering the need for replacement (Figure 7).

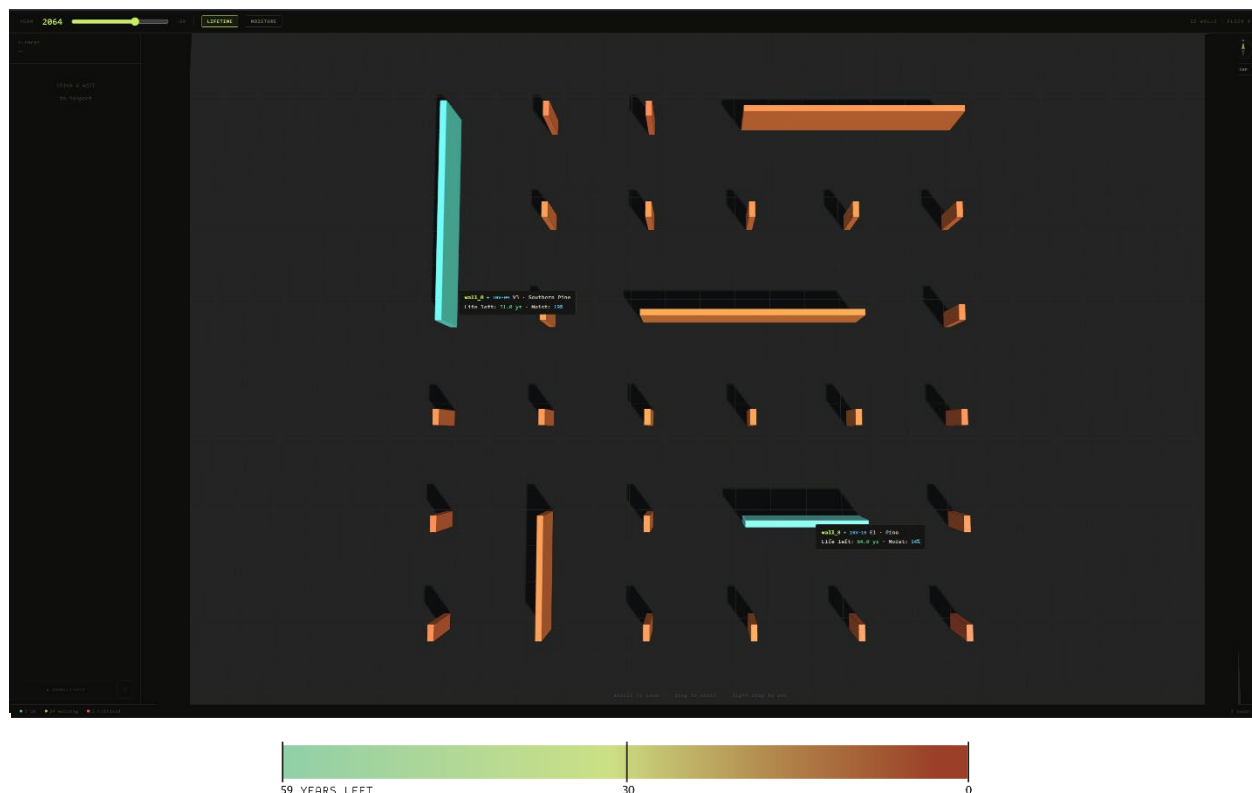


Figure 7. t(1) Substitution moment, year 40, wall_0 and wall_8 ID number elements

Through this setup, the model evaluates how targeted substitutions, guided by the substitution matrix and moisture-informed lifespan logic, can reduce temporal mismatch and contribute to a circular material economy by enabling the continued use of elements within their viable lifespan ranges rather than defaulting to replacement or disposal. The results therefore illustrate the capacity of the system to recalibrate element lifespans at both local and global scales, moving the building toward a more aligned temporal configuration. This minimizes both premature element replacement and underutilization of material lifespan.

3. Results

The demonstration study evaluates the system's capacity to identify and resolve temporal mismatches between building lifespan and material lifespans through element-level substitution. A single-story building model is analyzed over a simulated time horizon, during which elements exhibit differentiated lifespan decay based on their material properties and spatial exposure conditions. As the building approaches a 40-year operational threshold, a subset of elements falls below the target lifespan, producing a condition in which building lifespan exceeds the remaining service life of its components ($BL > ML$).

The system identifies these elements and enables their substitution with alternatives from the material inventory that exhibits higher residual capacity and more stable exposure conditions. These substitutions are constrained by application-specific rules, ensuring that replacement elements maintain or improve required performance criteria. Through this process, the overall

distribution of material lifespans within the building is recalibrated without altering the underlying design configuration.

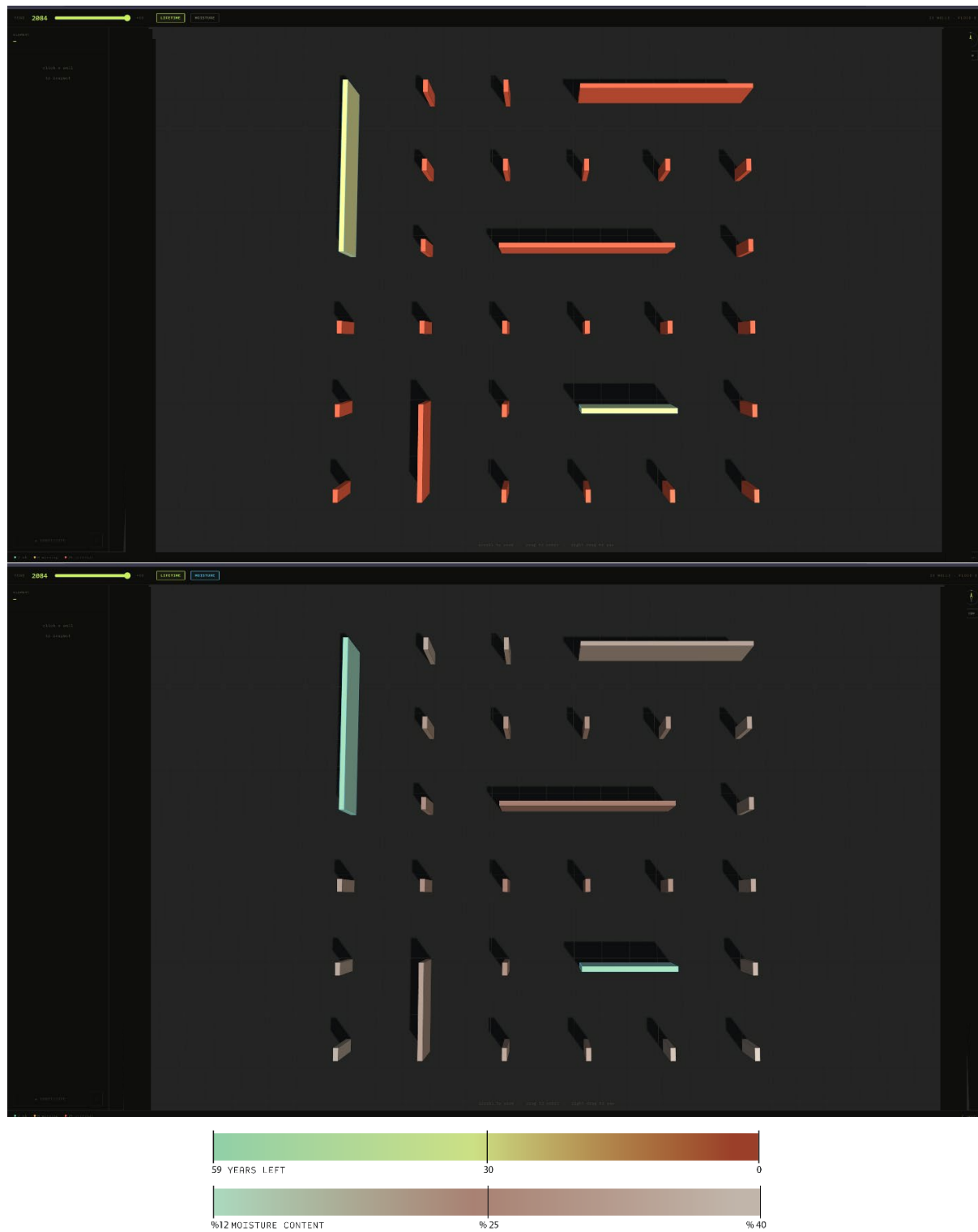


Figure 8. (t2) 40 years later, moisture exposure overlay (top), lifespan left (bottom)

The results demonstrate that temporal mismatch between building and material lifespans can be reduced through targeted reallocation of components. Elements that would otherwise require premature replacement are substituted with higher-performing alternatives, increasing the alignment between component-level and building-level service life. This enables the continued use of materials within their viable lifespan ranges and reduces reliance on new material input (Figure 8).

At the building scale, the system operates as a dynamic inventory model, in which elements are evaluated not only by type or specification, but by their remaining lifespan in relation to the overall assembly. This allows the design to adapt over time through discrete interventions, supporting iterative cycles of reuse, replacement, and reconfiguration. The demonstration study therefore illustrates how temporal data can inform material allocation decisions, shifting the design process from static specification toward lifespan-aware configuration.

4. Conclusions

This research introduces time as an operative parameter within architectural design, extending circular construction approaches by integrating temporal material behavior into decision-making processes. By encoding lifespan as a variable attribute at the level of individual elements, the framework enables a shift from static material specification to dynamic, inventory-based design. In doing so, it reframes the relationship between parts and wholes, allowing building lifespans to be aligned with the temporal capacities of their constituent materials.

The results suggest that temporal mismatch between building and material lifespans is not a fixed constraint, but a condition that can be actively managed through design. Rather than extending building life through new material input, the system reallocates existing elements based on residual capacity and exposure conditions, supporting more effective use of available material stock. This positions reuse not as a secondary consideration, but as a primary driver of design logic, where material selection and placement are informed by lifespan compatibility rather than uniform specification.

Future work will extend the framework toward more comprehensive temporal modeling and decision-making. This includes the development of optimization methods for automated material allocation, integration of additional performance criteria beyond moisture and lifespan, and the incorporation of upstream material timelines such as growth, harvesting, and processing. These extensions would enable the system to operate across broader spatial and temporal scales, supporting more complex and responsive design environments.

Table 2: Parameter mapping matrix

A. Forest parameters → Wood anatomy		B. Wood anatomy → Lumber properties		C. Lumber → CLT performance	
Forest condition	Biological effect in tree	Tree anatomy	Lumber behavior	Manufacturing stage	Added parameter
Light	ring width	Ring width	stiffness (MOE)	Drying	residual stress
Shade	slow growth	Density	strength (Fb)	Planing	surface

Precipitation	cell size	Knots	tensile failure/grading	Adhesive type	bonding quality creep + fire performance
Wind	reaction wood	Grain angle	shear failure	Press pressure	delamination resistance
Competition	knot frequency	Moisture	shrinkage warping	& Finger joints	tensile capacity
Soil nutrients	density	Latewood %	compression strength	CNC cutting	local stress concentrations
Temperature	latewood/earlywood ratio	Slope of grain	direction of wood fibers	Transport moisture	swelling

Beyond its immediate application within computational design, this work suggests broader implications across disciplines that shape and govern the built environment. Embedding temporal material data into design processes introduces new forms of knowledge about material performance over time, which may inform adjacent domains such as insurance, material warranties, salvage and reuse industries, and construction finance. In these contexts, decisions are often based on assumed lifespans or standardized risk models; the integration of time-dependent material data offers the potential for more precise, performance-based evaluations. By incorporating temporal reasoning into design, the role of the architect may extend toward participation in systems that manage risk, value, and long-term material stewardship, aligning architectural practice with disciplines concerned with lifecycle governance of the built environment.

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